RESEARCH REPORT

Healthcare Robotics: Precision, Recovery, and Care

A Study of Surgical, Rehabilitation, and Hospital-Support Robotics

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1. Introduction

Say the word “robot” in a hospital and most people still picture a science-fiction surgeon-bot doing an entire operation on its own. The reality is both less dramatic and more interesting. Today’s healthcare robots are tools that extend what surgeons, therapists, and nurses can already do — steadier hands in the operating room, tireless partners in physical therapy, and quiet helpers moving supplies through crowded hallways. None of them replace a clinician’s judgment; all of them are designed to sit alongside it.

The scale of this shift is real. The global surgical robots market alone was valued at roughly USD 11.3 billion in 2025 and is projected to grow at a compound annual rate of around 13 percent through the early 2030s, while the broader medical exoskeleton and rehabilitation robotics segment, though still much smaller in absolute terms, is growing even faster as hospitals look for ways to manage stroke recovery and an aging patient population. What ties all of this together is a shared set of problems: surgeons need more precision than the human hand alone can offer, therapists are stretched too thin to give every patient hours of one-on-one attention, and hospitals are short-staffed in ways that routine logistics work makes worse.

This report looks at three areas of healthcare robotics — surgical robots, rehabilitation and assistive devices, and the quieter category of hospital-support robots — and closes with the technologies, ethical questions, and future direction they all share.

2. Surgical Robots

Robotic surgery is the most mature and most visible branch of medical robotics, built almost entirely on the model Intuitive Surgical pioneered with its da Vinci system in the early 2000s. A surgeon sits at a console and controls robotic arms that translate hand movements into smaller, steadier motions inside the patient — filtering out natural hand tremor and allowing far finer movement than a scalpel held directly. Intuitive has placed more than 10,600 da Vinci systems in hospitals worldwide, and by the end of 2025 more than 17 million procedures had been performed on da Vinci systems since the platform launched. The company’s newest model, da Vinci 5, adds force feedback and enhanced 3D imaging, and in early 2026 received FDA approval for cardiac applications including mitral valve repair.

For most of the last two decades, Intuitive held something close to a monopoly in soft-tissue robotic surgery. That is changing quickly. Medtronic’s Hugo system and CMR Surgical’s Versius have both received regulatory clearance and are now being trialed at hospitals that had never adopted a da Vinci system, while Johnson & Johnson’s Ottava platform is moving through the FDA review process. Analysts covering the industry in 2026 describe this as the most significant competitive shift the field has seen since Intuitive itself entered the market.

2.1 Orthopedic and Specialized Surgical Robots

Surgery isn’t only about soft tissue. Stryker’s Mako system, used for knee and hip replacement, had assisted in more than 1.5 million procedures by 2025 and continues to expand into spine and shoulder work. These orthopedic platforms use pre-operative imaging to build a 3D model of a patient’s joint, then guide the surgeon’s instrument within a planned boundary during the procedure — a different approach from da

Vinci’s teleoperated arms, but built on the same underlying idea of combining a surgeon’s skill with a machine’s precision.

2.2 Access and Cost

None of this comes cheap. A surgical robot system typically costs between roughly USD 1.5 million and 2.5 million, with per-procedure disposable instruments adding several hundred to several thousand dollars on top. That price tag concentrates robotic surgery in well-funded hospitals in North America, Europe, and increasingly China and India, and raises a real question about which patients get access to it. It's also part of why hospital adoption in lower- and middle-income countries, including Pakistan, remains limited to a handful of major urban centers.

3. Rehabilitation and Assistive Robots

If surgical robots are about a single high-stakes moment, rehabilitation robots are about repetition — the unglamorous, time-consuming work of helping a patient's body relearn movement after a stroke, spinal cord injury, or surgery. The global rehabilitation robots market was valued at roughly USD 495 million in 2025 and is projected to grow at more than 15 percent annually through the mid-2030s, driven by rising stroke incidence and an aging population that outpaces the supply of physical therapists.

3.1 Exoskeletons

Wearable exoskeletons, made by companies such as Ekso Bionics, ReWalk Robotics, and Wandercraft, provide powered support to the legs or arms so a patient can stand, walk, or repeat a therapeutic movement with consistent, controlled assistance. These devices are increasingly common in stroke and spinal cord injury units: more than half of large hospitals in developed countries now use some form of robotic rehabilitation system, and the broader wearable exoskeleton market — spanning medical, industrial, and defense uses — is projected to grow from around USD 3.5 billion in 2026 to more than USD 60 billion by 2034. Researchers continue to refine the underlying interaction: a 2026 Northwestern University and Shirley Ryan AbilityLab study connected a therapist's and a patient's exoskeletons in real time, letting the therapist's movements guide the patient's gait training remotely.

3.2 Companion and Assistive Robots

Alongside physical rehabilitation, a smaller category of robots supports patients emotionally and cognitively. PARO, the robotic seal introduced in the earlier service-robots research, remains a well-known example for dementia care, and telepresence units let therapists or family members check in on patients remotely. These devices lean on the same computer-vision and sensor toolkit as exoskeletons and surgical robots, but are aimed at reducing isolation and monitoring wellbeing rather than restoring movement directly.

4. Hospital-Support Robots

Away from the operating table and the therapy room, a quieter category of robots handles the logistics that keep a hospital running — work that doesn't touch a patient directly but frees up staff time for the work that does.

4.1 Delivery and Logistics Robots

Autonomous mobile robots such as Aethon's TUG and Diligent Robotics' Moxi move medications, lab samples, linens, and supplies through hospital corridors, using onboard mapping to navigate elevators and hallways without dedicated tracks. Moxi is deployed in more than 25 U.S. hospitals and has completed over a million deliveries, and hospital staff at sites such as Cedars-Sinai have reported it cutting hundreds of miles of nurse walking within weeks of installation. TUG, meanwhile, is used in dozens of U.S. Veterans Affairs hospitals for similar tasks. The appeal is straightforward: routine transport is exactly the kind of repetitive task that pulls nurses away from patients, and automating it gives staff more time at the bedside.

4.2 Disinfection Robots

Ultraviolet-light disinfection robots, such as those made by Xenex and the UK's Akara Robotics, sanitize a hospital room in minutes by emitting UV-C light that damages the DNA of bacteria and viruses. Some hospitals using these systems have reported meaningful reductions in healthcare-associated infections, though the evidence varies by study and setting, and most experts describe UV robots as a useful addition to manual cleaning rather than a replacement for it.

5. Common Enabling Technologies

A da Vinci surgical arm, a rehabilitation exoskeleton, and a Moxi delivery robot look nothing alike, but they share a common toolkit. All rely on precise sensing — cameras, force sensors, and in the case of mobile robots, LiDAR — combined with some form of path planning or motion control software, and increasingly some layer of machine learning to adapt to a specific patient or environment. Cloud connectivity ties many of these systems to hospital record systems or remote monitoring dashboards, which is convenient but also introduces new points of failure: a 2024 security review found vulnerabilities in TUG's onboard systems, and hospitals are now expected to meet stricter connectivity standards before deploying similar robots. Advances in one category tend to spill into another, too — computer-vision techniques refined for surgical navigation have found their way into exoskeleton gait-tracking, and vice versa.

6. Challenges and Ethical Considerations

Healthcare robotics raises a distinct set of concerns compared to industrial or even domestic robots, largely because the stakes involve direct patient safety and highly personal data. Accountability is one open question: if a surgical robot malfunctions or an exoskeleton misjudges a patient's gait, it isn't always clear whether responsibility sits with the manufacturer, the hospital, or the clinician operating the device. Cost is another — the price of surgical systems and advanced exoskeletons puts them out of reach for many smaller hospitals and most facilities in lower-income countries, which risks widening rather than closing gaps in care quality. Data privacy matters here more than in most other robotics domains, since these systems often connect directly to electronic health records and continuously monitor patient movement or vital signs. And there's a softer concern that echoes across the whole field of care robotics: rehabilitation and companionship, in particular, involve a human relationship that a machine can support but arguably shouldn't fully replace.

7. Future Outlook

Every indicator points toward continued, fairly rapid growth. The surgical robots market is projected to more than triple by the mid-2030s as competition from Medtronic, Johnson & Johnson, and CMR Surgical pushes prices down and expands the pool of hospitals that can afford a system, while the rehabilitation robotics market is expected to grow even faster in percentage terms as stroke and spinal cord injury cases rise alongside an aging global population. Analysts expect AI-assisted workflow tools — software that helps a surgeon plan an approach or helps an exoskeleton adapt in real time to a specific patient's gait — to matter as much as the hardware itself over the next decade. For countries like Pakistan, where hospital capital budgets are tight and the ratio of physical therapists to patients is already low, the more realistic near-term opportunity probably lies in lower-cost categories: simple telepresence consultation setups, basic disinfection robots, and increasingly affordable modular exoskeletons rather than multimillion-dollar surgical platforms.

8. Conclusion

Healthcare robotics is not one technology but several, each solving a different problem within the same broader shortage of time, precision, and trained staff. Surgical robots give surgeons finer control over delicate procedures; rehabilitation robots give therapists a way to deliver more consistent, repeatable therapy than human hands alone can sustain over long sessions; and hospital-support robots quietly take routine transport and cleaning off overworked staff. None of these systems are a substitute for clinical judgment or human care — the evidence so far points toward partnership rather than replacement — but together they represent one of the clearer examples of robotics meaningfully changing how care is delivered, provided cost and accountability keep pace with the technology itself.

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